

Amanda Shum

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PROFESSIONAL SUMMARY

AI Engineer with experience in **Computer Vision, Deep Learning, Azure AI, workflow automation, and AI-assisted development** with additional experience in **technical product** and **project management**. Built CNN and object-detection systems using **PyTorch** and **TensorFlow**, and led cross-functional AI initiatives through product planning, stakeholder coordination, technical execution, and **Agile** delivery. Delivered measurable business impact, including 25% growth in rental bookings and a 75% reduction in administrative workload across 10,000+ inquiries.

EDUCATION

Simon Fraser University

Master of Professional Computer Science (Visual Computing)

Burnaby, BC

Expected Graduation Date: June 2027

Queen's University

Bachelor of Computing (Honours), Artificial Intelligence Specialization

Kingston, ON

June 2025

WORK EXPERIENCE

CanAI Garage, Technology and Strategy Branch (TSB) at Shared Services Canada

Vancouver, BC

Co-op Student

2026-Present

- Led the product and project delivery of a reusable GitHub Copilot customization framework with 20+ governed prompts, instructions, skills, and engineering standards, coordinating 2 to 4 rotating team members and establishing standardized context and evaluation practices to reduce repeated prompting, token usage, and time to usable outputs.
- Contribute to Azure-based AI and automation solutions, including RAG applications and public-sector workflow proposals using Azure OpenAI, Azure AI Document Intelligence, Azure AI Vision, Power Automate, and Power Apps for knowledge retrieval, document processing, email triage, and human-in-the-loop review.
- Develop technical documentation, governance models, workflow specifications, training materials, and user guidance while coordinating sprint deliverables through Azure DevOps and collaborating with stakeholders on architecture, privacy, scalability, Responsible AI, and operational readiness.

Kingston Student Housing Co-operative

Kingston, ON

Vice President

2021-2025

- Held operational and management roles within a student housing co-operative with 21 residential properties, overseeing operations, human resources, facilities, member services, safety compliance, and regulatory requirements while advising the Board of Directors on budgeting, policy, organizational planning, stakeholder management, software transitions, and revenue-generating initiatives.
- Increased rental bookings by 25% through customer analysis, seasonal marketing, and room-allocation strategies, and led Google Workspace and Google Apps Script automations that reduced administrative workload by 75% and supported responses to 10,000+ inquiries within 24 hours.

Queen's University

Kingston, ON

Teaching Assistant

2023-2025

- Supported undergraduate courses in Database Management Systems, Computer Architecture, and introductory Computer Science by delivering tutorials, grading assignments, managing course materials, facilitating forum discussions, and providing one-on-one guidance in SQL, relational database design, Python fundamentals, and problem-solving.

PROJECTS

Real-Time Pothole Detection and Relative Depth Estimation

Sept-Dec 2025

- Developed a **real-time pothole detection and relative-depth estimation pipeline** using *YOLOv8* and *MiDaS*, combining bounding-box region analysis, depth normalization, and noise filtering to distinguish potholes from surrounding road surfaces across varied lighting, textures, and camera perspectives.

KSHC Mailer and Workflow Automation System

2024-2025

- Designed and deployed an **automation system** for the Kingston Student Housing Cooperative using *Google Apps Script*, *Google Sheets*, and *HTML/JavaScript* for email communication, booking management, member applications, document generation, approval workflows, and role-based dashboards, improving data organization and reducing manual administrative processing.

Gridworld Coverage

Jan-Apr 2025

- Built and trained an RL agent using *Stable Baselines3*, *DQN*, *A2C*, and *PyTorch* in a custom Gridworld environment, optimizing reward functions and observation-space configurations to achieve an **85% area-coverage success rate**.

Fruit Ripeness Detection

Sept-Dec 2024

- Developed a two-stage deep learning system using *TensorFlow* and *NumPy*, combining a primary CNN for fruit classification with fruit-specific CNN models for ripeness prediction and achieving **96% fruit-classification accuracy**.

SKILLS

Programming Languages: Python, C, C++, JavaScript, PHP, SQL, R, HTML, CSS

Machine Learning and Computer Vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Ultralytics YOLO, Stable Baselines3, NumPy, pandas

AI and Cloud Technologies: Azure OpenAI, Azure AI Vision, Azure AI Document Intelligence, Retrieval-Augmented Generation, AWS, Power Automate, Power Apps

Development and Collaboration: Git, GitHub, Azure DevOps, Azure Boards, Visual Studio Code, Google Apps Script, SharePoint, Power BI

Core Competencies: Computer Vision, Image Processing, Deep Learning, Object Detection, Image Classification, Monocular Depth Estimation, Reinforcement Learning, Prompt Engineering, Context Engineering, Workflow Automation, Solution Architecture, Responsible AI, Agile Delivery, Product Planning, Project Coordination, Technical Documentation, Stakeholder Management

Additional Tools: Microsoft Office, Google Workspace, Microsoft Teams, Canva, Onshape, AnyDesk, GitHub Desktop, WordPress, Unity

Operating Systems: Windows, Linux